MATH 564

Begin with the Classifier concept. Suppose we have known data consisting of m vectors in $\mathbb{R}^p$ (feature vectors, x_k) along with corresponding class values ($y_k \in S$). That is, we have

$$\{(x_k, y_k)\}_{k=1}^{m}, \quad x_k \in \mathbb{R}^r, \quad y_k \in S$$

We seek a classifier function $f: \mathbb{R}^r \to \mathbb{R}$ and possibly $g: \mathbb{R} \to S$ so that $y_k \approx g \circ f(x_k)$ for all k. Then we can use f to classify new data points, x_j for which y_j is not known.

Before we jump to Neural Network classifiers, consider some effective and efficient alternatives.

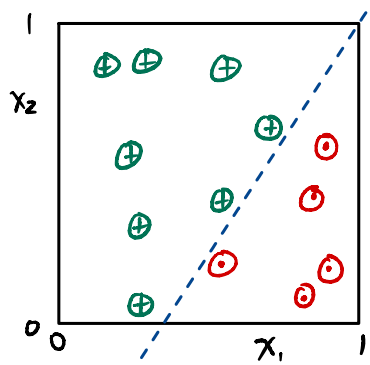
① Linear classifier. (Example in $\mathbb{R}^2$, but applicable in $\mathbb{R}^n$)

Divide known data into classes:

$\{(x_k, y_k)\}$ for $\oplus$

$\{(x_\ell, y_\ell)\}$ for $\odot$

Classifier $f(x) = -\frac{1}{2} + \frac{3}{2}x_1 - x_2$

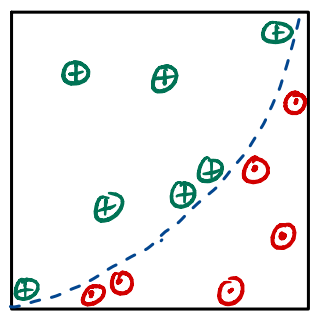


If $f(x) > 0$ then x is in the set of $\odot$
If $f(x) < 0$ then x is in the set of $\oplus$

② Nonlinear Classifier.

$f(x) = x_1^2 - x_2$

works as long as the data is actually separable by a function



Implementation: $f(x;w)$ is parameterized by a parameter vector w. For example:

$$f(x;w) = \sin w_1 x - \frac{x + w_2}{w_3}$$

we have the optimization problem

$$\max_{w} \min_{k} \left(f(x_{k,1}, w) - x_{k,2}\right)^2$$
$$\text{s.t.} \quad w \in \mathbb{R}^p$$

③ Linear Programming Formulation of ②:

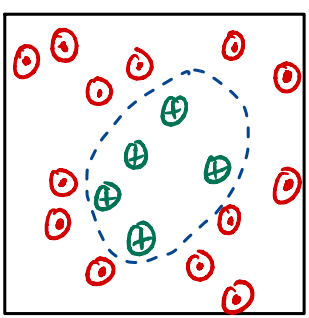
$$\max \quad \delta$$
$$\text{s.t.} \quad f(x_{k,1}; w) - y_k \geq \delta \quad \leftarrow \ominus$$
$$f(x_{\ell,1}; w) - y_\ell \geq \delta \quad \leftarrow \oplus$$
$$w \in \mathbb{R}^p$$
$$\delta \in \mathbb{R}$$

where $f(x;w) = \sum_{j=1}^{m} w_j \phi(x)$ ← linear combination

weights

fixed functions

④ Change of coordinates

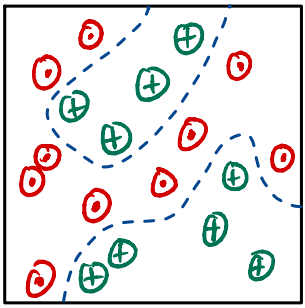


separable with radial functions!

In general: hypersperical coordinates. ugh!

These functions need not be smooth nor continuous nor even have a closed form expression. They might be from code or lookup table!

⑤ Level set concept approach



Raise the problem to one higher dimension so that

$f(x_k; w) > 0$ for class ⊙

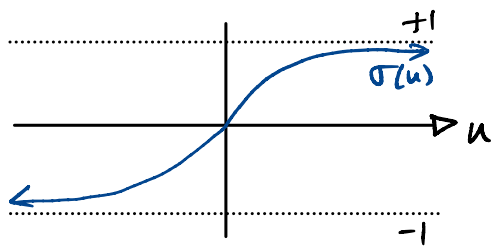
$f(x_\ell; w) < 0$ for class ⊕

- - - Is the zero level curve of f

In this case we could assign $y_k \in \{-1, +1\}$ then we seek f:

$$f(x; w) : \mathbb{R}^p \to \mathbb{R}$$

$$g(x; w) = \sigma \circ f \qquad \sigma = \frac{1-e^{-u}}{1+e^{-u}}$$



Then we have optimization problem

$$\min \; F(w) = \frac{1}{2}\sum_{k=1}^{m}\left(g(x_k; w) - y_k\right)^2$$

$$\text{s.t.} \quad w \in \mathbb{R}^p$$

Again, for some user-specified function $f(x; w)$

Another option: $\sigma = \dfrac{1}{1+e^{-u}}$ with $y_k \in \{0, 1\}$.

or even $y_k \in \{\Delta, 1-\Delta\}$ for more numerical stability.

Generalized to $y_k \in \{\Delta, 2\Delta, 3\Delta, \dots, 1-\Delta\}$, $\Delta = \frac{1}{c+1}$ for c ordered classes.

we will need $\nabla F(w)$ to employ our optimization methods.

$$F(w) = \frac{1}{2} \| g(x;w) - y \|^2$$

$$\nabla F(w) = (g(x;w) - y)^T \nabla g(x;w)$$

$$\nabla g(x;w) = \sigma'(f) \nabla f(x;w) = \sigma(f)(1-\sigma(f)) \nabla f(x;w)$$

(we used the curious and useful fact that $\sigma'(u) = \sigma(u)(1-\sigma(u))$ for $\sigma(u) = \frac{1}{1+e^{-u}}$.)

⑥ Feed Forward Neural Network.

* If no functional form for $f(x;w)$ is available or reasonable based on the given data and scenario.

matrix operations

$$g(x;w) = \sigma(W_d \sigma(\cdots \sigma(W_2 \sigma(W_1 x)) \cdots))$$

sigmoid function (above)

Notice that if σ = Identity, then $g = Wx$ for some matrix W (linear!). All nonlinearity is provided by nonlinear σ. Note also the complexity freedom given by matrix sequence $W_d W_{d-1} \cdots W_2 W_1$ that can be of any size as long as W_1 has r columns and W_d has one row

The computation is a sequence of similar operations:

$$L_1 = \sigma(W_1 x)$$
$$L_2 = \sigma(W_2 L_1)$$
$$\vdots$$
$$L_d = \sigma(W_d L_{d-1}) \quad (= g)$$
$$F(w) = \frac{1}{2} \| L_d - y \|^2$$